

ψ DAG: Projected Stochastic Approximation Iteration for Linear DAG Structure Learning

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Klea Ziu

Department of Machine Learning
Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, UAE



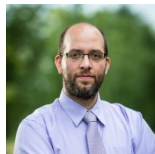
Slavomír Hanzely



Loka Li



Kun Zhang



Martin Takáč



Dmitry Kamzolov

Outline

- 1 Introduction
- 2 The ψ DAG Framework
- 3 Theoretical Guarantees
- 4 Results
- 5 Conclusions

Correlation $\neq$ Causation

Ice Cream Sales vs. Shark Attacks

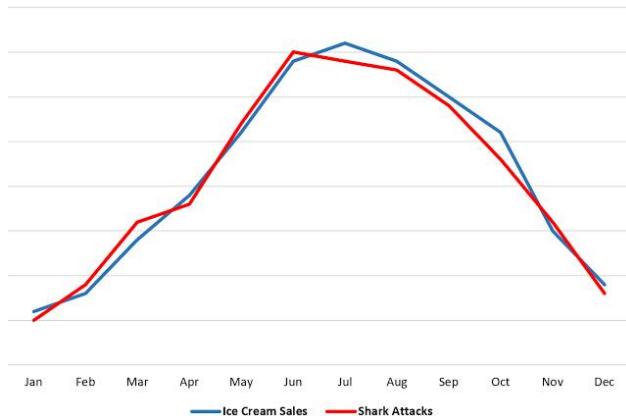


Figure: Ice cream sales vs. shark attacks.

Observation:

- Higher ice cream sales $\leftrightarrow$ more shark attacks
- Does ice cream *cause* shark attacks?

Correlation $\neq$ Causation

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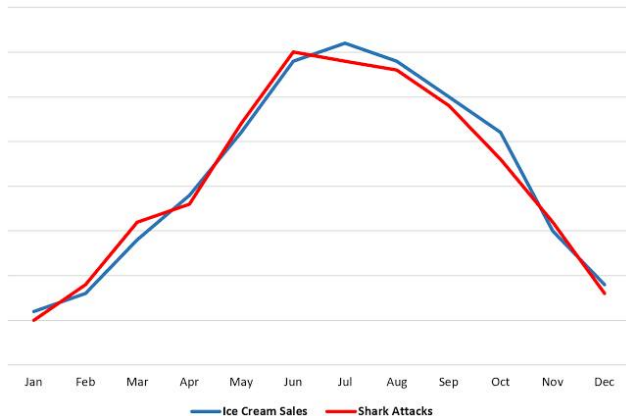


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Hidden cause: Hot weather

- More swimming $\rightarrow$ more shark exposure
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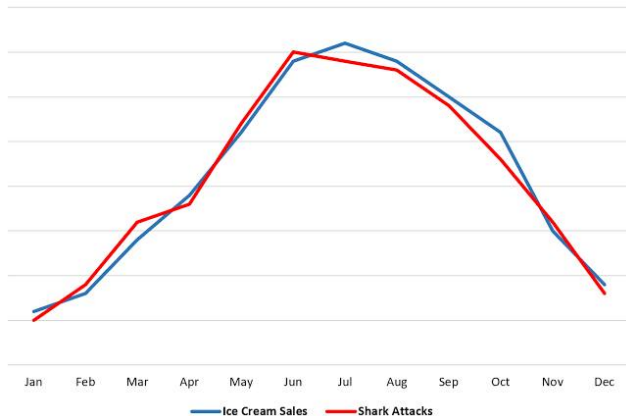


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Takeaway: Correlations can be misleading. We need a framework (DAGs) to reason about causality.

What is a Directed Acyclic Graph (DAG)?

Definition:

- **Directed:** Edges have direction ($X \rightarrow Y$)
- **Acyclic:** No cycles allowed
- **Graph:** Nodes connected by edges
- Directed edges show causal relationships

Applications:

- **Genomics:** Gene regulatory networks
- **Finance:** Market dependencies
- **Healthcare:** Treatment pathways

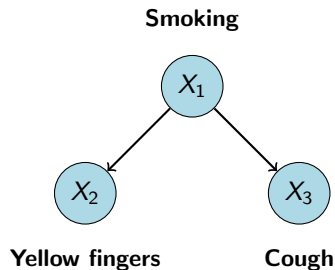


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Exponential search space: 2^{d^2-d} possible graphs

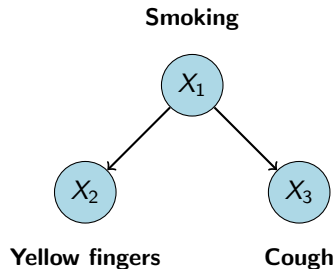


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NP-hard optimization problem!

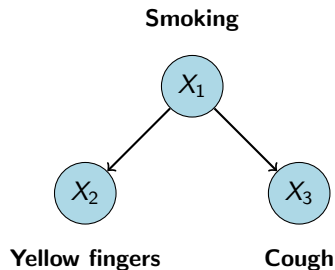


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Our Goal: Scale to 10,000+ variables efficiently

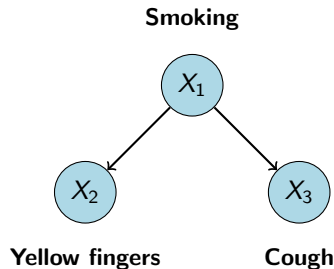
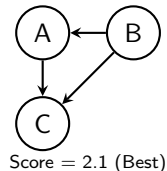
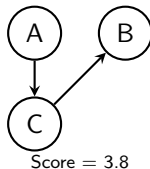
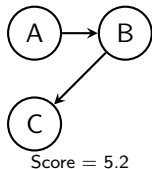


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What Are Score-Based Methods?

Intuition:

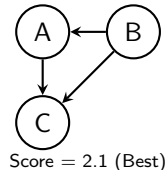
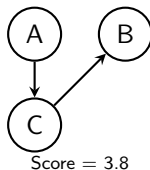
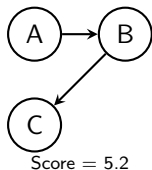
- We want to learn a Directed Acyclic Graph (DAG) that explains the data.
- A "score" measures how well a candidate graph fits the observed data.
- The best graph = the one with the lowest score (best fit).



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Key Point

Score-based methods = **fit data by minimizing a loss function**, while ensuring the graph is acyclic.

How Is the Score Calculated?

Linear DAG Model:

$$\mathbf{X} = \mathbf{XW} + \mathbf{N} \quad (1)$$

- **W**: weighted adjacency matrix of the DAG
- **N**: independent noise terms
- Graph structure determined from non-zero entries in **W**

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Classical Objective:

$$l(\mathbf{W}; \mathbf{X}) = \frac{1}{2n} \|\mathbf{X} - \mathbf{XW}\|_F^2 \quad (2)$$

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Connection

Methods like NOTEARS, GOLEM, NOCURL, DAGMA all aim to minimize this same score.

Limitations of Score-Based Methods

Challenges:

- High computational complexity: $O(d^3)$ per iteration
- Poor scalability beyond ~ 1000 nodes
- Numerical instabilities
- Convergence to stationary points

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Motivation

We need scalable and stable alternatives to classical score-based methods.

Key Innovation: Stochastic Reformulation

Sample Average Approximation (SAA)

$$\min_{\mathbf{W}} \frac{1}{n} \sum_{i=1}^n f(\mathbf{W}, X_i)$$

- Deterministic problem
- Solution depends on fixed sample size n
- Gap from true solution: δ_x, δ_F

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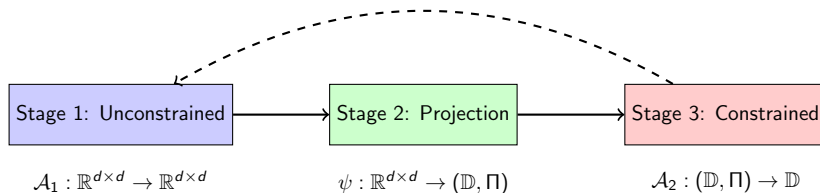
Stochastic Approximation (SA)

$$\min_{\mathbf{W} \in \mathcal{W}} \mathbb{E}_{x \sim \mathcal{D}} \left[\frac{1}{2} \|x - \mathbf{W}^T x\|^2 \right]$$

- Uses stochastic gradients
- No fixed sample size required
- Targets the true DAG in expectation

Advantages of SA: lower complexity, better scalability, targets true DAG in expectation

The Three-Stage Framework



- ① **Unconstrained:** SGD without DAG constraints
- ② **Projection:** Find closest DAG and topological ordering
- ③ **Constrained:** Optimize while preserving vertex order

Algorithm: ψ DAG

Algorithm 1 ψ DAG

```
1: Input: Initial model  $\mathbf{W}_0 \in \mathbb{R}^{d \times d}$ 
2: for  $k = 0, 1, 2, \dots, K - 1$  do
3:    $\mathbf{W}_k^{(1/3)} = \text{SGD}(\mathbf{W}_k)$                                      // Stage 1: Unconstrained
4:    $(\mathbf{W}_k^{(2/3)}, \pi_k) = \psi(\mathbf{W}_k^{(1/3)})$                        // Stage 2: Projection
5:    $\mathbf{W}_{k+1} = \text{SGD}_{\pi_k}(\mathbf{W}_k^{(2/3)})$                        // Stage 3: Constrained
6: end for
7: Output:  $\mathbf{W}_K$ 
```

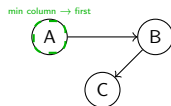
Key Features:

- Stage 1: exploration, Stage 2–3: DAG projection and constrained optimization
- Low computational complexity: $O(d^2)$ per iteration of SGD + projection
- Guarantees DAG output

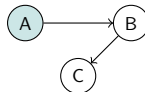
Efficient Projection Method: Step-by-Step

Closest DAG via Topological Sorting:

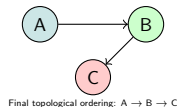
Step 1: Compute norms and select vertex



Step 2: Assign vertex to ordering



Step 3: Recursively sort remaining vertices



Final topological ordering: $A \rightarrow B \rightarrow C$

Complexity: Time $O(d^2)$, Space $O(d^2)$, Parallelizable

Theorem (Convergence Rate)

For L_1 -smooth function with σ_1 -stochastic gradients:

$$\mathbb{E}[F(\mathbf{W}_T) - F^*] \leq \mathcal{O}\left(\frac{\sigma_1 R}{\sqrt{T}} + \frac{L_1 R^2}{T}\right)$$

Key Properties:

- Converges to local minimum
- **DAG constraints satisfied**
- Recovers true DAG under correct ordering

Experimental Results (SHD)

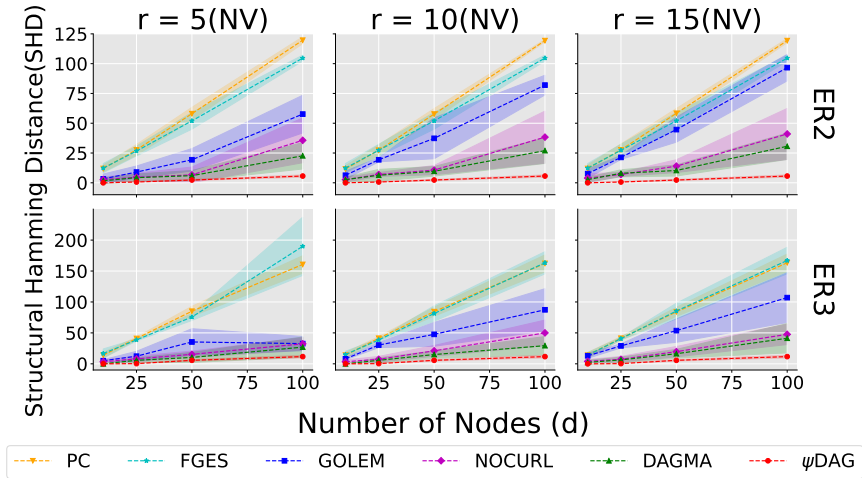


Figure: SHD

Experimental Results (Precision)

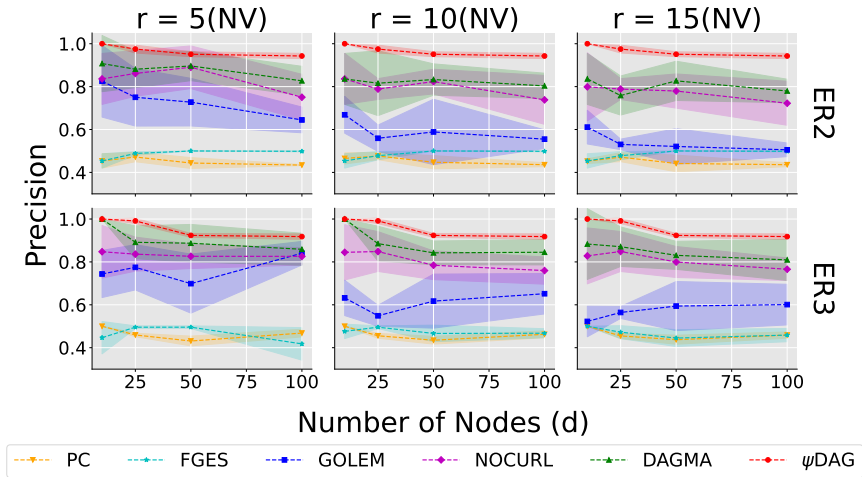


Figure: Structure Recovery under Gaussian noise.

Experimental Results (Scalability)

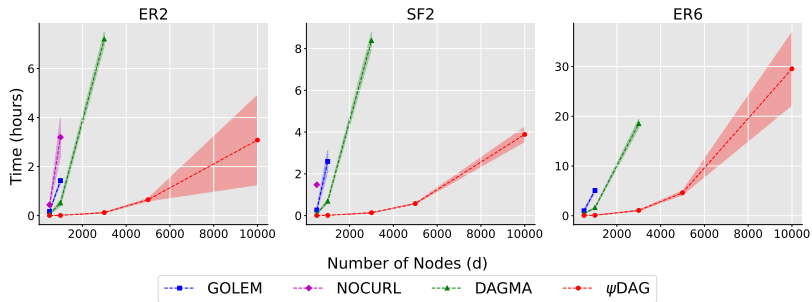


Figure: Runtime for large graphs.

Experimental Results (Scalability)

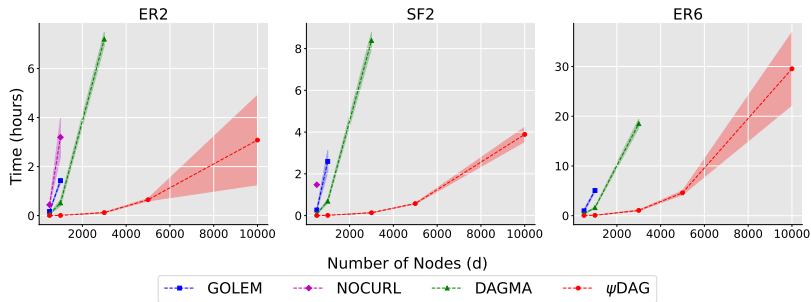


Figure: Runtime for large graphs.

Observations:

- 10-100x faster on large graphs
- Only method handling 10K nodes
- Linear scaling with graph size

Why ψ DAG Works?

Design Advantages:

- No matrix exponentials
- No hyperparameter tuning
- Guaranteed DAG output
- Parallelizable operations
- Memory efficient

Comparison:

- NOTEARS: $O(d^3)$ complexity
- GOLEM: No acyclicity guarantee
- DAGMA: Numerical instabilities
- ψ DAG: $O(d^2)$ complexity

Key Innovation

Stochastic optimization + Efficient projection = Scalability

Main Contributions

- 1 **Novel reformulation:** Stochastic optimization perspective
- 2 **Efficient algorithm:** ψ DAG with $O(d^2)$ complexity
- 3 **Scalability:** First method to handle 10,000+ nodes
- 4 **Robustness:** Superior performance under noise

Impact:

- Enables large-scale causal discovery
- Practical for real-world applications
- Opens new research directions

Future Directions: Extension to nonlinear SEMs

References



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Thank You!

Questions?

Contact: klea.ziu@mbzuai.ac.ae

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